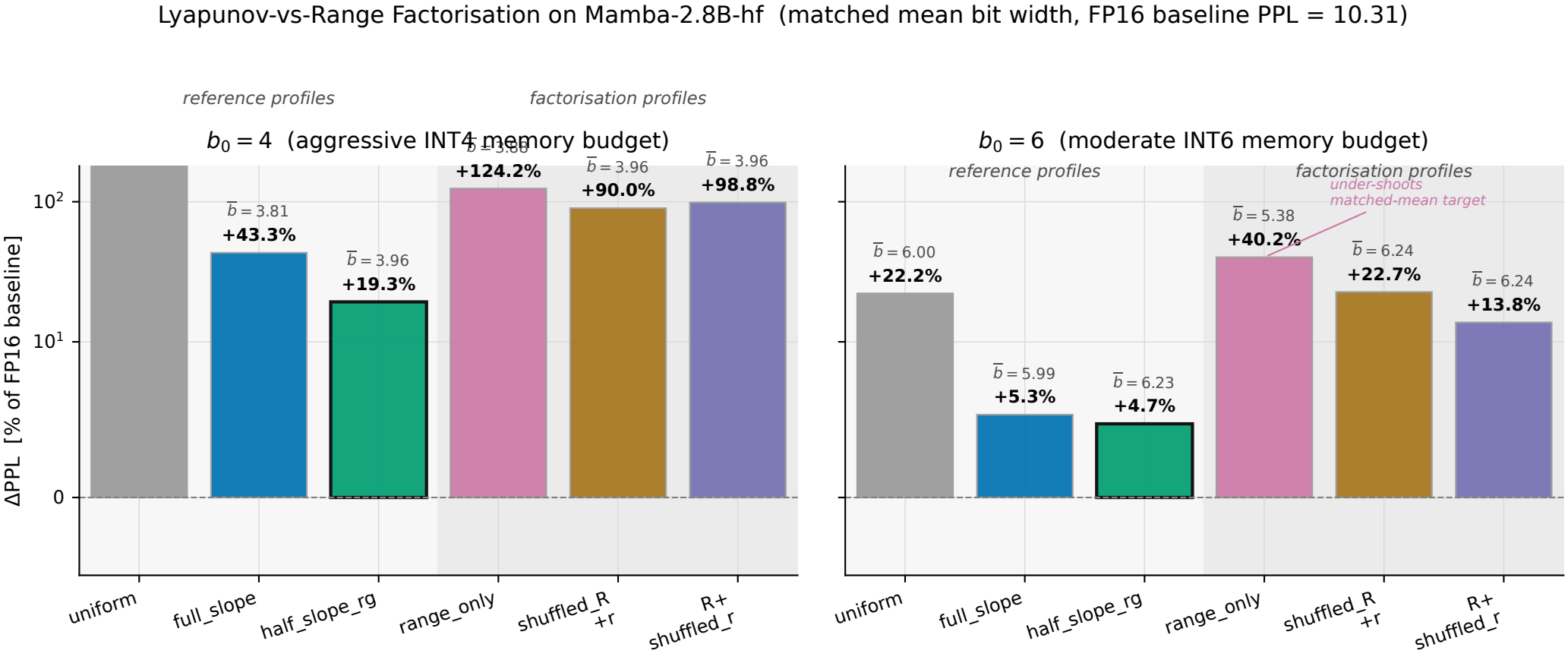
Lyapunov-vs-Range Factorisation on Mamba-2.8B-hf (matched mean bit width, FP16 baseline PPL = 10.31)
reference profiles
factorisation profiles
$b_0 = 4$ (aggressive INT4 memory budget)
$b_0 = 6$ (moderate INT6 memory budget)
ΔPPL [% of FP16 baseline]
10^2
10^1
0
$\overline{b} = 3.81$
+43.3%
$\overline{b} = 3.96$
+19.3%
$\overline{b} = 3.88$
+124.2%
$\overline{b} = 3.96$
+90.0%
$\overline{b} = 3.96$
+98.8%
$\overline{b} = 6.00$
+22.2%
$\overline{b} = 5.99$
+5.3%
$\overline{b} = 6.23$
+4.7%
$\overline{b} = 5.38$
+40.2%
$\overline{b} = 6.24$
+22.7%
$\overline{b} = 6.24$
+13.8%
under-shoots
matched-mean target
uniform
full_slope
half_slope_rg
range_only
shuffled_R +r
R+ shuffled_r